An extreme solitary wave for the bidirectional Whitham system

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Abstract

We construct an extreme solitary wave for the bidirectional Whitham system. Its velocity profile is positive, even, continuous, smooth away from its crest and strictly decreasing to zero on the positive half-line. The crest reaches the first critical value of the cubic traveling-wave equation. The construction starts with positive highest periodic waves of arbitrarily large period. A common modulus of continuity and a fixed drop from the critical height keep their limit nonconstant. A balance identity using the kernel's first absolute moment then excludes every positive limiting pedestal, without assuming an integrable profile.

1 Introduction

The bidirectional Whitham system is the nonlocal dispersive model

$$\partial_t \eta + \partial_x (K * v + \eta v) = 0, \qquad \partial_t v + \partial_x \left(\eta + \frac{1}{2} v^2 \right) = 0, \tag{1}$$

where η is the surface elevation, v is the velocity and the convolution kernel on $\mathbb{R}$ has symbol

$$\widehat{K}(\xi) = \frac{\tanh \xi}{\xi}, \qquad \widehat{K}(0) = 1. \tag{2}$$

It combines the finite-depth gravity-wave dispersion relation with a quadratic shallow-water nonlinearity; see [2, 6]. We study the largest crest allowed by its scalar traveling-wave equation.

For profiles moving at speed c and tending to zero at infinity, integration of (1) gives $\eta = cv - v^2/2$ and

$$K * v = c^2 v - \frac{3c}{2} v^2 + \frac{1}{2} v^3. \tag{3}$$

The derivative of the cubic first vanishes at $v = c(1 - 1/\sqrt{3})$. At this height the smooth inversion used for smaller waves degenerates. The existence question is therefore whether a localized wave can attain this crest while retaining positive, monotone tails.

Small solitary waves for this system are supplied by the variational construction of Nilsson and Wang [6, Theorem 3.1]. Highest periodic waves were constructed by Ehrnström, Johnson and Claassen [2]; their Appendix B discusses positive highest periodic waves at the fixed period of that paper. Ehrnström, Mæhlen and Varholm [3, Section 4.2] determine precise crest behavior for extreme solutions and explicitly leave existence of a highest solitary wave open in Remark 4.8. A periodic highest wave alone does not answer that question: as its period grows, it might approach a constant, or retain a positive value at infinity.

The following result supplies a solitary profile at the critical crest.

Theorem 1.1. *There exist*

$$1 < c < \sqrt{3(\sqrt{3}-1)}$$

and an even continuous function $v : \mathbb{R} \to (0,\infty)$, smooth on $\mathbb{R} \setminus \{0\}$, such that v is strictly decreasing on $(0,\infty)$, $\lim_{|x|\to\infty} v(x) = 0$, and

$$K * v = c^2 v - \frac{3c}{2}v^2 + \frac{1}{2}v^3 \quad \text{on } \mathbb{R}, \qquad v(0) = c\left(1 - \frac{1}{\sqrt{3}}\right).$$

The surface profile $\eta = cv - v^2/2$ satisfies $\eta(0) = c^2/3$. The pair $(\eta(x-ct), v(x-ct))$ solves (1) in distributions, and classically away from the moving crest.

The theorem gives an affirmative answer to the existence question in [3, Remark 4.8], for the exact symbol and conservative system (1). It is an existence result for one limiting speed; uniqueness, stability and a family indexed by all speeds are separate questions.

We obtain the profile as a limit of highest positive periodic waves. Periodic-to-solitary constructions are available for the unidirectional Whitham equation; Ehrnström, Nik and Walker [4] give a direct construction of a full family. That equation has a different multiplier and a quadratic profile nonlinearity, and its treatment of nonzero tails uses a Galilean transformation. Here the cubic retains the speed in several coefficients. We instead integrate the convolution defect over expanding symmetric intervals. The first absolute moment of K makes this defect tend to zero for any bounded profile with equal tails. Strict concavity of the cubic contradicts that balance if the limiting tail is positive.

Two estimates allow the period to grow. A kernel-translation estimate gives one modulus of continuity for all heights up to the critical value, and a differentiated equation for smooth periodic waves gives a drop from that height on a fixed interval. Taking highest limits preserves the drop without differentiating at a crest. We also construct the positive periodic branch for every sufficiently large period, using the corrected analytic continuation theorem of [1]; this step is needed beyond the fixed-period positive-wave discussion in [2].

Section 2 gives the kernel facts and the uniform estimates. Section 3 constructs the positive highest periodic waves, and Section 4 passes to the real line and proves the theorem. Throughout, $\widehat{f}(\xi) = \int_{\mathbb{R}} e^{-ix\xi} f(x)\,\mathrm{d}x$, with inverse factor $1/(2\pi)$, and convolution uses Lebesgue measure.

2 The kernel and estimates uniform in the period

Write

$$a = 1 - \frac{1}{\sqrt{3}}, \qquad \gamma(c) = ac, \qquad F_c(t) = c^2 t - \frac{3c}{2}t^2 + \frac{1}{2}t^3, \qquad c_\gamma = \sqrt{3(\sqrt{3}-1)}. \tag{4}$$

The first positive critical point of the cubic is $\gamma(c)$. In fact,

$$F_c'(t) = \frac{3}{2}(\gamma(c) - t)\left(\gamma(c) - t + \frac{2c}{\sqrt{3}}\right), \qquad F_c''(t) = 3(t-c). \tag{5}$$

Thus F_c is strictly increasing and strictly concave on $[0,\gamma(c)]$ when $c > 0$. A profile with maximum $\gamma(c)$ will be called highest. Smooth profiles lying strictly below this value will be called subcritical; this refers to their height, rather than their speed.

The normalized kernel is

$$K(x) = \frac{1}{\pi}\log\coth\frac{\pi|x|}{4}, \qquad x \neq 0. \tag{6}$$

This is the formula in [3, Section 4.2] for the symbol used here. It fixes the scalar in the convolution. The qualitative kernel results of [2, Lemma 2.2] also apply. In particular, K is even, positive and integrable, with mass one, and

$$K'(x) = -\frac{1}{2\sinh(\pi x/2)}, \qquad x > 0. \tag{7}$$

The logarithmic singularity is integrable. All derivatives decay exponentially away from zero, and $\int_{\mathbb{R}} |x|K(x)\,\mathrm{d}x < \infty$.

For a physical period $P > 0$, set $k = 2\pi/P$ and

$$K_P(x) = \sum_{j\in\mathbb{Z}} K(x + jP). \tag{8}$$

The sum converges in L^1 on a period and has mass one. Its differentiated sums converge uniformly on compact sets avoiding $P\mathbb{Z}$. It is strictly decreasing and strictly convex on $(0, P/2)$. One way to see the strict signs for every P is to use

$$K(x) = \frac{2}{\pi}\sum_{j=0}^{\infty} \frac{e^{-(2j+1)\pi|x|/2}}{2j+1}.$$

Each periodized exponential is a positive multiple of $e^{-\mu x} + e^{-\mu(P-x)}$ on $0 < x < P$, and has the stated signs on the half-period.

On writing $\theta = kx$, convolution becomes the even 2π-periodic Fourier multiplier

$$\mathcal{K}_k \cos(n\theta) = m(kn)\cos(n\theta), \qquad m(\xi) = \frac{\tanh\xi}{\xi}, \quad m(0) = 1. \tag{9}$$

For each fixed $k > 0$, this symbol has order -1. The periodic multiplier theorem [2, Proposition 2.7] therefore gives $\mathcal{K}_k : C^s \to C^{s+1}$ at positive noninteger exponents, understood via the equivalent Hölder–Zygmund spaces. We use the Banach space

$$X = C^{1/4}_{\mathrm{even}}(\mathbb{T}), \qquad \|u\|_{C^\alpha} = \|u\|_\infty + \sup_{x\neq y} \frac{|u(x) - u(y)|}{d_{\mathbb{T}}(x, y)^\alpha}.$$

Consequently $\mathcal{K}_k : X \to X$ is compact. These mapping bounds may depend on P.

We first record a strict comparison that also applies to continuous highest profiles.

Lemma 2.1. *If a continuous, even, nonconstant P-periodic function u is nonincreasing on $[0, P/2]$, then $K_P * u$ is strictly decreasing on $(0, P/2)$. In particular, a solution of $K_P * u = F_c(u)$ taking values in $[0, \gamma(c)]$ is strictly decreasing there.*

Proof. Subtract $\min u$ and write the resulting nonnegative function as the integral of its superlevel indicators. For every level strictly between its minimum and maximum, the superlevel set is a centered interval of radius $r \in (0, P/2)$, up to endpoints. For such an interval,

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(K_P * \mathbf{1}_{[-r,r]}\big)(x) = K_P(x + r) - K_P(x - r) < 0$$

for almost every $x \in (0, P/2)$. Indeed the circle distance of $x + r$ is strictly larger than $|x - r|$, also when $x + r > P/2$. The interval convolution is locally absolutely continuous, since K_P is integrable. It is therefore strictly decreasing on every subinterval. Tonelli's theorem and integration over the nonempty interval of levels preserve strict decrease. The last assertion follows from the strict increase of F_c on its stated range. □

Lemma 2.2 (A common modulus of continuity)**.** *Let $1 \le c \le c_\gamma$ and let v be a bounded solution of $K * v = F_c(v)$ on $\mathbb{R}$, with $0 \le v \le \gamma(c)$. Periodic solutions are allowed. With $M = ac_\gamma$, one has*

$$|v(x+h) - v(x)|^2 \le \frac{2M}{\sqrt{3}} \|K(\cdot + h) - K\|_{L^1(\mathbb{R})}. \tag{10}$$

These profiles have a common $C^\beta(\mathbb{R})$ bound for every fixed $0 < \beta < 1/2$.

Proof. For $0 \le s < t \le \gamma(c)$, (5) gives

$$F_c(t) - F_c(s) \ge \sqrt{3}c \int_s^t (\gamma(c) - z)\,\mathrm{d}z \ge \frac{\sqrt{3}}{2}(t-s)^2.$$

The last integral, at fixed $t - s$, is smallest when $t = \gamma(c)$. The equation and $\|v\|_\infty \le M$ now imply (10). Evenness and decrease of K imply, for $h > 0$,

$$\|K(\cdot + h) - K\|_1 = 4\int_0^{h/2} K(t)\,\mathrm{d}t \le Ch(1 + |\log h|), \qquad 0 < h \le \tfrac{1}{2}. \tag{11}$$

The positive and negative parts change sign at $-h/2$, which gives the equality. The estimate follows from (6); the case $h < 0$ follows by translation. Combining these bounds proves the claim. □

The estimate requires no crest equality. For each fixed period it gives a bounded subset of $C^{3/8}$, hence precompactness in $X = C^{1/4}_{\text{even}}$. The latter embedding follows from uniform convergence and the interpolation estimate $[f]_{C^\alpha} \le 2\|f\|_\infty d^{-\alpha} + [f]_{C^\beta} d^{\beta-\alpha}$, for $\beta > \alpha$.

Lemma 2.3 (Speed and small constant states)**.** *A nonconstant positive periodic solution with $v < \gamma(c)$ and $c > 0$ has $1 < c < c_\gamma$. At $c = 1$, the only nonnegative continuous periodic solution with $0 \le v \le \gamma(1)$ is zero. At $c = c_\gamma$, the nonnegative continuous periodic solutions with $0 \le v \le \gamma(c_\gamma)$ are the constants zero and $\gamma(c_\gamma)$.*

For each fixed P, all solutions sufficiently near $(v, c) = (0, 1)$ in $X \times \mathbb{R}$ are constant. At every $c_* > 1$, zero has a neighborhood containing no nonconstant solutions.

Proof. Integrating over a period gives $\int (F_c(v) - v) = 0$. On $0 < t \le \gamma(c)$ the factor

$$\frac{F_c(t) - t}{t} = c^2 - 1 - \frac{3c}{2}t + \frac{1}{2}t^2$$

is strictly negative for $0 < c \le 1$. For $c \ge c_\gamma$ it is nonnegative, because it decreases up to $\gamma(c)$ and

$$\frac{F_c(\gamma(c))}{\gamma(c)} = \frac{c^2(1+\sqrt{3})}{6} \ge 1. \tag{12}$$

At $c = c_\gamma$ equality occurs only at $t = \gamma(c)$ or, before division, at $t = 0$. Continuity gives the constant alternatives.

For the assertion near $(0, 1)$, write $v = s + h$, where s is its mean and h has mean zero. Project $\mathcal{K}_k v - F_c(v) = 0$ onto mean-zero functions. The derivative in h at $(s, h, c) = (0, 0, 1)$ is $\mathcal{K}_k - I$. Its kernel on that subspace is zero, since $m(kn) < 1$ for all $n \ne 0$. As a compact perturbation of $-I$, it is Fredholm of index zero and is invertible. The analytic implicit-function theorem, for example [5, Lemma 3R], gives $h = 0$ as the unique small solution of the projected equation for each nearby (s, c). The full equation is thus the scalar constant equation. Its nearby roots are zero and

$$b(c) = \frac{3c - \sqrt{c^2 + 8}}{2}. \tag{13}$$

At zero and $c_* > 1$, the full derivative is $\mathcal{K}_k - c_*^2 I$. Convolution has norm at most one for the full Hölder norm, so the Neumann series gives its inverse. The same implicit-function argument proves the last assertion. All neighborhoods here may depend on P. □

Lemma 2.4 (Strict nodal signs)**.** *A smooth, even, nonconstant periodic solution with $0 < v < \gamma(c)$ that is nonincreasing on $[0, P/2]$ satisfies*

$$v'(x) < 0 \quad (0 < x < P/2), \qquad v''(0) < 0, \qquad v''(P/2) > 0.$$

This property is open relative to the subcritical solution set in $X \times \mathbb{R}$.

Proof. The reflected derivative identity is

$$F_c'(v(x))v'(x) = \int_0^{P/2} [K_P(x-y) - K_P(x+y)]v'(y)\,\mathrm{d}y. \tag{14}$$

The bracket is positive inside the half-period. Since $v' \le 0$ and is negative on some interval, the right side is strictly negative. At the endpoints, differentiation and one integration by parts give

$$F_c'(v(0))v''(0) = -2\int_0^{P/2} K_P'(y)v'(y)\,\mathrm{d}y < 0,$$

and

$$F_c'(v(P/2))v''(P/2) = 2\int_0^{P/2} K_P'(P/2-y)v'(y)\,\mathrm{d}y > 0.$$

The integrals converge because v' vanishes linearly at the endpoints, whereas K_P' has at most an inverse-distance singularity. This is the nodal argument of [2, Theorem 4.2], with physical period P.

Near any subcritical solution there is a positive gap from $\gamma(c)$. The identity $v = F_c^{-1}(\mathcal{K}_k v)$ and the one-derivative gain improve convergence of solutions in X to convergence in C^2. Interior derivative signs and the endpoint second-derivative signs are then preserved. □

The next estimate prevents the highest waves from becoming flat when the period grows.

Proposition 2.5 (A uniform drop from the critical height)**.** *There are $r_0 > 0$ and $D_0 > 0$, independent of $P \ge 4$, such that every smooth nonconstant solution as in Lemma 2.4, with $1 < c < c_\gamma$, satisfies*

$$\gamma(c) - v(r) \ge D_0 r, \qquad 0 < r \le r_0. \tag{15}$$

The estimate is inherited by its uniform limits.

Proof. The image terms in $K_P'(t) - K'(t)$ are uniformly bounded for $P \ge 4$ and $0 < t \le 1/2$, by exponential decay. Since $-K'(t) \sim 1/(\pi t)$, we can choose r_0 with $2r_0 < 1/2$ so that

$$-K_P'(2r) \ge \frac{1}{4\pi r}, \qquad 0 < r \le r_0, \quad P \ge 4. \tag{16}$$

On the negative half-period the profile is strictly increasing. For $x, y \in [-r, -r/2]$, convexity of K_P gives

$$K_P(x-y) - K_P(x+y) = \int_{|x-y|}^{|x+y|} (-K_P'(t))\,\mathrm{d}t \ge r[-K_P'(2r)].$$

The improper nonnegative integral has the same interpretation if the lower endpoint is zero. Reflecting (14) to this half-period, restricting its positive integral to $[-r, -r/2]$, and using the decrease of F_c' yield

$$F_c'(v(-r))v'(x) \ge r[-K_P'(2r)][v(-r/2) - v(-r)], \qquad -r < x < -r/2.$$

Integrating in x and cancelling the strictly positive profile increment gives $F_c'(v(-r)) \geq r/(8\pi)$. Put $d = \gamma(c) - v(-r)$. From (5),

$$F_c'(v(-r)) = \sqrt{3}c\,d + \frac{3}{2}d^2 \leq Bd, \qquad B = c_\gamma\left(\sqrt{3} + \frac{3a}{2}\right).$$

Thus (15) holds with $D_0 = 1/(8\pi B)$. Only smooth subcritical profiles have been differentiated. The final inequality passes to uniform limits directly. □

3 Positive highest periodic waves

The positive constant branch (13) satisfies $0 < b(c) < \gamma(c)$ for $1 < c < c_\gamma$. We construct a nonconstant branch from it, keeping the physical period fixed throughout this section.

Lemma 3.1 (Local bifurcation from the positive constant)**.** *For every sufficiently large P, there is a unique $c_b \in (1, c_\gamma)$ such that a local analytic branch of even solutions bifurcates from $(b(c_b), c_b)$ in the first cosine mode. Its positive-amplitude arm has positive profiles with the nodal signs of Lemma 2.4. The speed is nonconstant, and the profile linearization is invertible at nearby nonzero points of this branch.*

Proof. Set $v = b(c) + w$ and $\Lambda(c) = F_c'(b(c))$. With $s(c) = \sqrt{c^2/4 + 2}$, direct calculation gives

$$\Lambda(c) = \frac{c^2}{4} + 3 - \frac{3}{2}cs(c), \qquad \Lambda'(c) = \frac{c}{2} - \frac{3}{2}\left(s(c) + \frac{c^2}{4s(c)}\right) < 0. \tag{17}$$

It decreases from 1 to 0 on $[1, c_\gamma]$. Since $0 < m(k) < 1$, there is exactly one c_b with $\Lambda(c_b) = m(k)$. At that point $\mathcal{K}_k - \Lambda(c_b)I$ has even kernel spanned by $\cos\theta$. It is Fredholm of index zero, and its range is precisely the functions with zero first cosine coefficient. The nonzero derivative in (17) gives transversality. The analytic simple-bifurcation theorem in [1, Remark 3] gives a local curve containing all nearby nonconstant solutions. Parametrize it by its first cosine coefficient τ. Half-period translation changes τ to $-\tau$, so local uniqueness makes the speed even in τ.

Let $A(c) = 3(b(c) - c)/2$. The equation for w is

$$(\mathcal{K}_k - \Lambda(c))w = A(c)w^2 + \frac{1}{2}w^3. \tag{18}$$

Analyticity gives expansions, for this fixed P,

$$w = \tau\cos\theta + \tau^2(w_0 + w_2\cos 2\theta) + O(\tau^3), \qquad c = c_b + C_2\tau^2 + O(\tau^4).$$

All coefficients on the next line are evaluated at c_b:

$$\begin{aligned} w_0 &= \frac{A}{2(1 - m(k))}, \qquad w_2 = \frac{A}{2(m(2k) - m(k))}, \\ -\Lambda' C_2 &= A^2\left(\frac{1}{1 - m(k)} + \frac{1}{2(m(2k) - m(k))}\right) + \frac{3}{8}. \end{aligned} \tag{19}$$

The constant and second modes at order two give w_0, w_2; the first mode at order three gives $-\Lambda' C_2 = 2Aw_0 + Aw_2 + 3/8$. As $k \downarrow 0$,

$$1 - m(k) = \frac{k^2}{3} + O(k^4), \qquad m(k) - m(2k) = k^2 + O(k^4), \qquad A \longrightarrow -\frac{3}{2}, \quad \Lambda' \longrightarrow -2.$$

Consequently

$$k^2 C_2 \longrightarrow \frac{45}{16} > 0. \tag{20}$$

For sufficiently large P this coefficient is nonzero. The local Lyapunov–Schmidt equation has the form $G(c,\tau) = \tau(\ell(c-c_b) + q\tau^2 + \cdots)$, with $\ell \neq 0$ and $q \neq 0$. At its nonzero solutions, differentiation in τ with c fixed gives $G_\tau = 2q\tau^2 + o(\tau^2)$. This is the profile Schur complement, so the full profile linearization is invertible there.

The local curve and its expansion can also be constructed in $C^{2+1/4}$; the same simple kernel and transversality apply, and uniqueness in X identifies the curves. The leading term $b(c) + \tau\cos\theta$ therefore gives positivity and the strict nodal pattern for $\tau > 0$ small. □

We use the corrected analytic global-bifurcation theorem of Constantin, Strauss and Varvaruca [1, Theorem 4, Remarks 3–4]. For a real analytic map $\mathcal{F} : \mathcal{O} \subset \mathbb{R} \times X \to X$, its hypotheses are: the full trivial line lies in $\mathcal{O}$; one point on it has simple kernel and cokernel with transverse parameter crossing; the profile derivative is Fredholm of index zero at every solution; and $\mathcal{O}$ is exhausted by bounded closed sets whose solution subsets are compact. The distinguished analytic continuation of the local branch then either eventually leaves every set of that exhaustion or is a periodic curve. At each branch point, its incoming distinguished analytic arc has a uniquely selected outgoing arc for which the joined curve admits a local uniformizing real-analytic parameter [1, Remark 3 and Figure 2, preprint pp. 14–15]. This parameter is one-to-one on a sufficiently small interval around the join. At regular points the corresponding continuation is the unique implicit-function graph. These are local properties of the selected arc. We use compact-exhaustion escape, rather than the stronger norm/boundary disjunction corrected in that paper.

Proposition 3.2 (Global continuation and monotonicity)**.** *For every sufficiently large fixed P, the positive local arm in Lemma 3.1 extends to a distinguished global curve of positive smooth subcritical profiles. All of its positive-side profiles are nonconstant and strictly decreasing on $(0, P/2)$. The curve leaves every compact exhaustion set of the positive subcritical solution domain.*

Proof. To place the whole positive constant branch on the trivial line, use the analytic speed parametrization

$$c(\lambda) = 1 + (c_\gamma - 1)\frac{e^\lambda}{1+e^\lambda}, \qquad \lambda \in \mathbb{R},$$

and define

$$\mathcal{F}(\lambda, w) = \mathcal{K}_k w - \left[F_{c(\lambda)}(b(c(\lambda)) + w) - F_{c(\lambda)}(b(c(\lambda)))\right]$$

on the open set

$$\mathcal{O} = \{(\lambda, w) \in \mathbb{R} \times X : \min v > 0,\ \max v < \gamma(c(\lambda)), \quad v = b(c(\lambda)) + w\}.$$

The whole $w = 0$ line belongs to this set. The local crossing is simple and transverse by Lemma 3.1, since $c'(\lambda) > 0$. At every solution the profile derivative is $\mathcal{K}_k - F_c'(v)$. Multiplication by $F_c'(v) > 0$ is invertible on X, whereas $\mathcal{K}_k$ is compact. The derivative is therefore Fredholm of index zero.

For integer $j \geq 1$, take the closed subsets of $\mathcal{O}$ given by

$$|\lambda|, \|w\|_X \leq j, \qquad \min v \geq 1/j, \qquad \gamma(c) - \max v \geq 1/j.$$

They exhaust $\mathcal{O}$. On each of them, the inverse cubic has uniformly bounded derivatives on the relevant range. The equation and the one-derivative gain give a uniform $C^{1+1/4}$

bound for its solutions. Their solution subsets are compact in X and closed. This verifies all hypotheses of the continuation theorem.

It remains to keep the chosen arm in the monotone cone. The cone of even nonincreasing functions is closed in X. At any nonconstant solution in the cone, Lemma 2.4 makes the strict nodal property open relative to the solution curve. Its first loss at a finite parameter could therefore occur only at a constant. In $\mathcal{O}$ the only constant solution is $b(c)$.

At such a contact the profile derivative must be singular, or the implicit-function theorem would allow only the constant branch nearby. Its kernel is $\cos(n\theta)$ for exactly one $n \geq 1$. The transverse local bifurcation theorem applies there too. For $n \geq 2$, neither sign of the leading cosine is monotone on $[0, \pi]$, which excludes such a contact. For $n = 1$, strict decrease of Λ identifies the contact with the original bifurcation point.

For completeness, the remaining same-arm return is excluded as follows. Suppose the first return is at a parameter $T > 0$. Arrival along the negative local arm violates monotonicity already before T. Arrival along the positive local arm identifies its analytic germ with the outgoing arm near zero, with reversed orientation. Match these two copies by an order-reversing local parameter map $s \mapsto h(s)$. The distinguished arc continuation in [1, Figure 2] extends this match through each regular point and each intervening branch point. At a regular point this follows from uniqueness of the implicit-function graph; at a branch point it follows from the unique selection of the outgoing arc just stated. Continue while $0 < s < h(s) < T$. The left parameter increases and the right one decreases. If continuation stopped at two distinct interior limits, the local uniformizing charts at those limits and their transition map would extend the match. Thus they meet at some $s_* = h(s_*)$. The two sides at that point would then describe the identical germ, contrary to local uniformizing injectivity. This rules out the return. The nearby regular points needed for the distinguished arc construction are supplied by (20) and the Schur complement in Lemma 3.1.

Hence the entire positive side remains nonconstant and monotone. A periodic continuation would return to its initial constant point, and is also excluded. The remaining global alternative is escape from every compact exhaustion set. □

Proposition 3.3 (Highest waves for large periods)**.** *For every sufficiently large P, there are $c_P \in (1, c_\gamma)$ and a positive, even, continuous P-periodic solution v_P with*

$$K * v_P = F_{c_P}(v_P), \qquad v_P(0) = \gamma(c_P).$$

It is strictly decreasing on $(0, P/2)$, smooth off $P\mathbb{Z}$, and satisfies (15) and the common modulus (10).

Proof. Fix P and follow the positive side of Proposition 3.2. Lemma 2.2 makes all its profiles bounded in $C^{3/8}$ and precompact in X. Its speeds stay away from 1: otherwise an X-convergent subsequence has limiting speed 1, its limiting profile is zero by Lemma 2.3, and the constant-only neighborhood there excludes the nonconstant sequence. Its speeds also stay away from c_γ. At that endpoint the limit must be zero or $\gamma(c_\gamma)$. Local invertibility at zero excludes the first, and the fixed positive drop (15) excludes the second.

The minima stay uniformly positive along this fixed-period curve. Indeed, a limiting zero minimum at an interior limiting speed would give a nonnegative solution vanishing at one point. Positivity of K_P then forces the limit to be identically zero; local invertibility at that speed again excludes it.

If the crest gap also stayed uniformly positive, the whole positive curve would lie in one of the exhaustion sets used above. This contradicts Proposition 3.2. There is therefore a sequence whose crest gaps tend to zero. Take an X-convergent subsequence, also convergent in speed. Its limit solves the equation and has crest $\gamma(c_P)$. Estimate (15)

makes it nonconstant. A zero would force the limit to vanish identically by positivity of convolution, so it is positive. It is strictly decreasing by Lemma 2.1. Wherever it is below its crest, the localized inverse-cubic smoothing of [2, Lemma 4.1], with kernel K_P, makes it smooth. Strict decrease places these points precisely off $P\mathbb{Z}$. Both uniform estimates pass to the limit. □

4 The solitary limit

Choose $P_n \to \infty$ and the highest profiles from Proposition 3.3. We keep the physical spatial coordinate x, so the uniform modulus does not change with n.

Lemma 4.1. *After passage to a subsequence, $c_{P_n} \to c \in [1, c_\gamma]$ and $v_{P_n} \to v$ locally uniformly on $\mathbb{R}$. The limit is even, continuous, nonnegative and nonincreasing on $(0, \infty)$, and*

$$K * v = F_c(v), \qquad v(0) = \gamma(c), \qquad \gamma(c) - v(r) \geq D_0 r \quad (0 < r \leq r_0). \tag{21}$$

In particular, it is nonzero and nonconstant. Its limits at both infinities agree with a number $b \geq 0$ satisfying $F_c(b) = b$.

Proof. The suprema are bounded by M and the modulus in Lemma 2.2 is common. Compact-interval selection and a diagonal subsequence give the stated local uniform convergence; the speed can be selected first from its compact interval. Evenness and monotonicity pass to the limit on every compact set. The crest and drop also pass directly.

For fixed x, split the convolution into $|y| \leq R$ and $|y| > R$. On the first part $v_{P_n}(x - y)$ converges uniformly; the complementary error is bounded by $2M \int_{|y|>R} K(y)\,\mathrm{d}y$. Taking $n \to \infty$ and then $R \to \infty$ proves the pointwise equation. The decreasing bounded positive-half-line profile has a finite limit b, also the negative tail by evenness. Dominated convergence in $K * v(x) = \int K(y)v(x - y)\,\mathrm{d}y$ as $x \to \infty$ yields $F_c(b) = b$. No global uniform convergence is needed. □

We next show that the tails are zero. The following balance only requires equal bounded tails, rather than an integrable profile.

Lemma 4.2 (First-moment balance)**.** *If f is bounded and has the same finite limit at both infinities, then*

$$\lim_{R\to\infty} \int_{-R}^{R} (K * f - f)(x)\,\mathrm{d}x = 0. \tag{22}$$

Proof. For each fixed R, the absolute double integral for $K * f$ is at most $2R \|f\|_\infty \|K\|_1$. Fubini's theorem and the mass-one identity give

$$\int_{-R}^{R} (K * f - f)(x)\,\mathrm{d}x = \int_{\mathbb{R}} K(y) A_R(y)\,\mathrm{d}y, \quad A_R(y) = \int_{-R-y}^{R-y} f(x)\,\mathrm{d}x - \int_{-R}^{R} f(x)\,\mathrm{d}x.$$

For fixed y, its two boundary integrals tend to equal and opposite limits, so $A_R(y) \to 0$. For all R, y, $|A_R(y)| \leq 2\|f\|_\infty |y|$. The finite first absolute moment of K supplies an integrable majorant, proving (22). □

Lemma 4.3. *The limit in Lemma 4.1 has $1 < c < c_\gamma$ and $b = 0$.*

Proof. If $c = 1$, the equation gives

$$K * v - v = -\frac{1}{2} v^2 (3 - v) \leq 0.$$

It is strictly negative on a nonempty open interval, since $v(0) = a > 0$ and $v \le a < 3$. Its integrals over expanding symmetric intervals cannot tend to zero, contrary to Lemma 4.2. If $c = c_\gamma$, then $F_c(\gamma(c)) = \gamma(c)$. At $x = 0$, however, the positive mass-one average $K * v$ is strictly smaller than $\gamma(c)$, because v is a nonconstant continuous function bounded above by $\gamma(c)$. Thus the speed is interior.

Suppose $b > 0$. The drop in (21) gives $b < \gamma(c)$. Since $F_c(b) = b$,

$$F_c'(b) = 1 - \frac{3}{2}cb + b^2 < 1.$$

Put $w = v - b \ge 0$. It is nonzero, bounded and has zero tails. Strict concavity of F_c on $[b, \gamma(c)]$ gives

$$K * w = F_c(b + w) - F_c(b) < F_c'(b)w < w$$

wherever $w > 0$, with equality in the first difference when $w = 0$. Hence $K * w - w$ is nonpositive everywhere and strictly negative on an open interval. Lemma 4.2 applied to w is again a contradiction. This proves $b = 0$. □

Lemma 4.4. *The limiting profile is strictly positive on $\mathbb{R}$, strictly decreasing on $(0, \infty)$, and smooth on $\mathbb{R} \setminus \{0\}$.*

Proof. A zero of v would give $K * v > 0$ at that point, because K is positive and v is nonnegative and nonzero. This contradicts $F_c(0) = 0$.

For each level $0 < t < v(0)$, the superlevel set of v is a centered interval of finite positive radius $r(t)$, up to endpoints. Its finiteness uses the zero tail. For every $r > 0$, the function $K * \mathbf{1}_{[-r,r]}$ is locally absolutely continuous and has derivative

$$K(x + r) - K(x - r) < 0$$

for almost every $x > 0$. The logarithmic singularity at $x = r$ is integrable. Each such interval convolution is strictly decreasing. Integrating over levels by Tonelli's theorem shows that $K * v$ is strictly decreasing. Strict increase of F_c on $[0, \gamma(c)]$ then gives strict decrease of v. In particular $v(x) < \gamma(c)$ for $x \ne 0$.

For smoothness, convolution gains one Sobolev derivative: $K : H^s(\mathbb{R}) \to H^{s+1}(\mathbb{R})$, since $\sqrt{1 + \xi^2}\, m(\xi)$ is bounded. On a compact interval away from zero, localize v with a smooth cutoff equal to one on a slightly larger interval. The localized part is initially in L^2, so its convolution is in H^1. The complementary convolution is smooth on the smaller interval, since its kernel argument stays away from zero and every off-origin derivative is integrable at infinity against bounded v. On that interval $F_c'(v)$ has a positive lower bound. Its smooth scalar inverse, extended off the local range with bounded derivatives, therefore gives $v \in H^1$ locally by the weak chain rule.

Repeat with nested cutoffs. If v is locally in H^j, convolution gains one derivative. Composition with the smooth local inverse preserves this integer regularity: in one dimension the lower derivatives in each chain-rule term are bounded by the local H^1 embedding, while at most one highest derivative is only in L^2. Smooth approximation and the weak product and chain rules justify the calculation. Iteration and Sobolev embedding yield smoothness off zero. □

Proof of Theorem 1.1. Lemmas 4.1, 4.3 and 4.4 give all assertions about c and v. Set $\eta = cv - v^2/2$. Expanding the cubic equation yields the exact identities

$$K * v + \eta v = c\eta, \qquad \eta + \frac{1}{2}v^2 = cv. \tag{23}$$

For any locally integrable profile f, the change of variables $z = x - ct$ in a smooth compactly supported space-time test integral gives $\partial_t f(x - ct) = -c\partial_x f(x - ct)$ in distributions. Convolution by K commutes with translation since $K \in L^1$ and v is bounded. Applying distributional derivatives to (23) therefore gives both equations of (1). Smoothness makes them classical off the moving crest. Finally $a - a^2/2 = 1/3$, so $\eta(0) = c^2/3$. □

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